

Fatema Tuj Johora

<https://github.com/JohoraAva>

Email : Fatema.TujJohora@utdallas.edu

[Homepage](#)

WORK EXPERIENCE

- **Graduate Teaching Assistant** Dallas, Texas
The University of Texas at Dallas *August.2025– Ongoing*
- **Software engineer** Dhaka, Bangladesh
Synesis IT *April.2025– July,2025*

EDUCATION

- **The University of Texas at Dallas** Dallas, Texas
Doctor of Philosophy(PhD) in Computer Science *Fall'25–Ongoing*
- **Bangladesh University of Engineering and Technology** Dhaka, Bangladesh
Bachelor of Engineering in Computer Science and Engineering; CGPA: 3.94 out of 4.00 *Feb.2020–Mar.2025*
- **Ideal School and College,Motijheel** Dhaka, Bangladesh
HSC,GPA: 5.00 out of 5.00; Talent Pool scholarship *2019*
- **BTCL Ideal School and College,Moghbazar** Dhaka, Bangladesh
SSC,GPA: 5.00 out of 5.00 *2017*

SKILLS AND INTERESTS

- **Imperative Programming Languages:**C++, Java,Python, SQL, JavaScript, Bash, HTML, CSS, React
Functional Programming Languages:RocQ prover, Proverif, Tamarin, nuXmv

RESEARCH EXPERIENCE

- **Uncovering Deviations Under Protocol State Fuzzing in TLS Implementations** Dec 2023 – Present
Undergraduate Thesis, Supervised by Dr. Md. Shohrab Hossain, Professor, CSE, BUET
Collaboration with Dr. Syed Rafiul Hussain and Syed Md. Mukit Rashid, Department of CSE, Pennsylvania State University, USA,
Description: Developed an extension to the StateLearner framework for TLS 1.3 in Java to systematically test a range of TLS libraries. Conducted a thorough analysis of the full TLS 1.3 handshake process to detect deviations, verify protocol compliance, and identify bugs within tested libraries.

ACADEMIC SERVICES

- **Reviewer**
 - IEEE Transactions on Networking, 2025 [External]

PROJECTS

- **Kidney Anomaly Detection with Deep Learning Model:** Developed as part of the CSE-472(Machine Learning Sessional) course in the 4-2 as a two person project. This project includes:
 - Utilized four different deep learning models (VGG19, modified Xception, InceptionV3, and DenseNet) to detect kidney anomalies such as stones, cysts, and tumors, each producing individual predictions.
 - Merged the predictions from all models to improve accuracy and implemented a Feedforward Neural Network (FNN) for further enhancement of detection performance.
- **PixelShare:** Developed as part of the CSE-408(Software Development Sessional) course in the 4-1 with a three-person team.
 - A picture-sharing platform that allows users to upload photos and videos to designated groups, featuring an efficient tag-based search system for easy media discovery.
 - Fully implemented using Svelte, ensuring a responsive and dynamic user experience while supporting organized sharing and accessibility for group-based content.

- **Sherlock's Eye:** Developed as part of the CSE-216(Database Sessional) course in the 2-2 with a two-person team.
 - Enables users to file cases and track progress, while detectives can select cases based on expertise for efficient resolution.
 - Features real-time updates, status tracking, and secure communication channels to enhance interactions.
 - Built with Node.js (backend) and HTML/CSS (frontend), focusing on usability and security.
- **Securus Domas: Automated Home and Security:** Developed as part of the CSE-316(Microprocessors, Microcontrollers, and Embedded Systems Sessional) course in the 3-1 with a six-person team. This project includes:
 - Bluetooth-controlled door and light system, along with gas and flame detection, providing real-time alerts via Telegram.
 - Intruder detection utilizing LDR and laser technology, complemented by automated water tank management and plant watering systems.
 - Integrated rain sensor for automatic window closure, ensuring optimal safety and resource management.
- **Football Player Database System:** Developed as part of the CSE-108(OOP Sessional) course in the 1-2 as a single person project. This project includes:
 - A system to manage football players in different clubs, making it easy to handle player profiles, club assignments, and transfer tracking.
 - Built with JavaFX for the user interface and a server-client network system for smooth data sharing and communication.

ACADEMIC ACHIEVEMENTS

Scholarships: Dean's List Scholarship (L1-T2, L2-T2), University Merit Scholarship (L2-T1, L4-T1)